

University Project
Medical Decision Analysis

Medical Decision Analysis

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Academic Year: 2025–2026

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1 Definition of Medical Decision Analysis

Medical Decision Analysis (MDA) is a formal, systematic, and quantitative framework designed to support decision-making in healthcare under conditions of uncertainty. It integrates clinical evidence, probability theory, utility theory, and patient preferences to evaluate and compare alternative medical strategies such as diagnostic tests, treatments, or preventive interventions.

In clinical practice, physicians are frequently confronted with complex decisions involving uncertain outcomes, multiple therapeutic options, and trade-offs between benefits, risks, costs, and quality of life. Medical decision analysis provides a structured methodology that allows these elements to be explicitly modeled and analyzed.

At its core, medical decision analysis aims to identify the decision that maximizes expected benefit, usually expressed in terms of expected utility or health outcomes such as Quality-Adjusted Life Years (QALYs). The approach does not replace clinical judgment but rather complements it by offering transparency, consistency, and rational justification for medical choices.

Medical decision analysis is particularly useful in the following contexts:

- When multiple medical alternatives exist
- When outcomes are probabilistic rather than deterministic
- When patient preferences strongly influence decision outcomes
- When ethical, economic, and clinical dimensions must be balanced

By combining quantitative rigor with patient-centered considerations, medical decision analysis contributes to evidence-based medicine and supports personalized healthcare decisions.

2 Necessary Concepts for an Objective Approach to Medical Decision

2.1 Decision Alternatives

Decision alternatives represent the set of possible actions available to the decision-maker. In medical contexts, these alternatives may include choosing between treatment options, performing or avoiding diagnostic tests, or selecting preventive strategies. Alternatives must be clearly defined, mutually exclusive, and collectively exhaustive to ensure a valid analysis.

2.2 States of Nature and Clinical Outcomes

States of nature correspond to the possible outcomes that may occur following a chosen alternative. These outcomes are beyond the direct control of the decision-maker and may include recovery, partial improvement, adverse effects, complications, or death. Each outcome must be precisely specified to avoid ambiguity.

2.3 Probability Assessment

Probabilities quantify uncertainty by assigning likelihoods to each possible outcome. These probabilities are typically derived from:

- Clinical trials
- Epidemiological studies
- Meta-analyses
- Expert medical opinion

Probabilities must satisfy the axioms of probability theory, namely that they lie between 0 and 1 and that the sum of probabilities over all possible outcomes equals 1.

2.4 Payoffs and Outcome Measurement

Payoffs represent the value associated with each outcome. In medical decision analysis, payoffs are not limited to monetary measures and often include health-related metrics such as survival time, symptom reduction, or quality of life. Common outcome measures include life expectancy, Quality-Adjusted Life Years (QALYs), and Disability-Adjusted Life Years (DALYs).

2.5 Utility Theory

Utility theory provides a formal representation of preferences under uncertainty. Utilities are numerical values that reflect the desirability of specific health states from the patient's perspective. Typically, utilities are normalized such that:

$$0 = \text{Death}, \quad 1 = \text{Perfect health}$$

Utilities can be elicited using methods such as the standard gamble, time trade-off, or visual analog scales.

2.6 Decision Trees

Decision trees are graphical representations used to model sequential decisions and chance events. They consist of decision nodes, chance nodes, and terminal nodes, allowing a clear visualization of the decision process and facilitating expected utility calculations.

2.7 Expected Utility Principle

The expected utility of a decision is calculated as the sum of the utilities of all possible outcomes, weighted by their probabilities:

$$EU = \sum_{i=1}^n P_i \times U_i$$

According to expected utility theory, the optimal decision is the one that maximizes expected utility.

3 Patient Decision Criteria and Utility Perspective

3.1 Patient-Centered Decision Making

Modern healthcare emphasizes patient-centered decision making, recognizing that patients may value outcomes differently based on personal beliefs, lifestyle, and social context. Incorporating patient preferences into decision models ensures that medical choices align with individual values.

3.2 Risk Attitudes

Patients differ in their attitudes toward risk. Some patients are risk-averse and prefer options with more predictable outcomes, while others may be risk-seeking or risk-neutral. These attitudes are captured through utility functions and directly influence decision outcomes.

3.3 Multi-Criteria Decision Considerations

Medical decisions often involve multiple criteria, including effectiveness, safety, cost, and quality of life. Multi-criteria decision analysis extends traditional frameworks by allowing explicit weighting of these criteria, providing a more comprehensive evaluation of alternatives.

3.4 Ethical Dimensions

Although medical decision analysis provides quantitative guidance, ethical principles remain fundamental. Decisions must respect patient autonomy, promote beneficence, avoid harm, and ensure fairness. Decision analysis should therefore be viewed as a supportive tool rather than a substitute for ethical and clinical judgment.

4 Case Study: Pima Indians Diabetes Dataset

To demonstrate the application of medical decision analysis in a practical context, we selected the **Pima Indians Diabetes Dataset** from the UCI Machine Learning Repository. This dataset is widely recognized in the field of medical decision analysis and provides real clinical measurements for diabetes prediction.

4.1 Dataset Description

The dataset includes clinical and demographic variables for female patients of Pima Indian heritage, aged 21 and older. Key features include:

- Number of pregnancies
- Plasma glucose concentration
- Blood pressure
- Skin thickness
- Insulin level
- Body mass index (BMI)
- Diabetes pedigree function
- Age

The target variable is **Outcome**, which indicates whether the patient has diabetes (1) or not (0). The dataset contains over 700 observations and has been used extensively in academic research for predictive modeling and decision analysis.

4.2 Probabilistic Modeling

A logistic regression model was trained to estimate the probability of diabetes for each patient. Logistic regression is suitable for binary outcomes and provides probabilistic predictions that can be integrated with expected utility calculations.

For a selected patient, the predicted probability of having diabetes is given by:

$$p = 0.267$$

indicating a relatively low risk for diabetes.

4.3 Utility-Based Decision Rule

Expected utilities were calculated for two possible decisions:

- **Early Treatment:** initiating preventive or therapeutic measures immediately
- **Monitoring Only:** regular follow-up and observation without immediate treatment

Using predefined utility values for each outcome, the expected utilities for the selected patient were computed as:

$$EU(\text{Early Treatment}) = 0.680$$

$$EU(\text{Monitoring Only}) = 0.777$$

According to the **expected utility principle**, the optimal decision is the one that maximizes expected utility. In this case:

$$EU(\text{Monitoring Only}) > EU(\text{Early Treatment})$$

Hence, **monitoring is the preferred strategy** for this patient.

4.4 Clinical Interpretation

The low predicted probability of diabetes suggests that initiating treatment could expose the patient to unnecessary side effects, psychological burden, and cost without substantial benefit. Monitoring preserves quality of life while remaining clinically safe, highlighting the patient-centered approach of medical decision analysis.

4.5 Academic Relevance

This case study integrates real clinical data, probabilistic modeling, and utility-based decision rules. It illustrates the application of medical decision analysis principles in practice, providing reproducible results, transparent methodology, and patient-centered recommendations suitable for academic and clinical contexts.